

0.1 Change of rings and localization

Notation. • Let R, R', R'' be commutative rings,

- $I \subseteq R, I' \subseteq R'$ ideals,
- $S \subseteq (R, 1, \cdot)$ a submonoid,
- $\varphi : R \rightarrow R'$ a ring homomorphism,
- M, N modules over R .

Definition 0.1. (a) The functor

$$\begin{aligned} R' \otimes_R - : {}_R\mathbf{Mod} &\rightarrow {}_{R'}\mathbf{Mod} \\ M &\mapsto R' \otimes_R M \end{aligned}$$

is called the *base change* or *scalar extension* from R to R' via φ .

(b) The functor

$$\begin{aligned} (-)_R : {}_{R'}\mathbf{Mod} &\rightarrow {}_R\mathbf{Mod} \\ M' = (M', 0', +', \cdot') &\mapsto M'_R = (M', 0', +', \cdot) \end{aligned}$$

with $r \cdot m' := \varphi(r) \cdot' m'$ for $r \in R, m' \in M'$ is called *scalar restriction* from R' to R .

(c) The functor

$$\begin{aligned} S^{-1}(-) : {}_R\mathbf{Mod} &\rightarrow {}_{S^{-1}R}\mathbf{Mod} \\ M &\mapsto S^{-1}M \end{aligned}$$

is called *localization* at S .

Remark. Note that (c) is a special case of (a) because $S^{-1}(-) = S^{-1}R \otimes_R -$ (for localization homomorphism $\varphi_S : R \rightarrow S^{-1}R, r \mapsto \frac{r}{1}$).

Definition. Call R' flat over R (via φ) : $\iff \varphi : R \rightarrow R'$ is flat : $\iff (R')_R$ is a flat R -module.

Proposition 0.2. (a) *Scalar extension and restriction are additive functors.*

(b) *Scalar restriction is exact.*

(c) *Scalar extension is right exact, and exact $\iff R'$ is flat over R .*

(d) *For $N' \in {}_{R'}\mathbf{Mod}$:*

$$N' \otimes_{R'} (R' \otimes_R M) \cong N'_R \otimes_R M$$

(e) *Let $M' := R' \otimes_R M$ and $N' := R' \otimes_R N$, then*

$$M' \otimes_{R'} N' \cong R' \otimes_R (M \otimes_R N) \cong M'_R \otimes_R N$$

special case:

$$S^{-1}M \otimes_{S^{-1}R} S^{-1}N \cong S^{-1}(M \otimes_R N)$$

Proposition 0.3. Let $\varphi_{S,M} : M \rightarrow S^{-1}M, m \mapsto \frac{m}{1}$ be the localization map, then:

- (a) $S^{-1}(-)$ is exact.
- (b) $\varphi_{S,R} : R \rightarrow S^{-1}R$ is flat.
- (c) The map

$$\begin{aligned} \{N \leq_R M\} &\xrightarrow{p} \{\mathcal{N} \leq_{S^{-1}R} S^{-1}M\} \\ N &\mapsto S^{-1}N \end{aligned}$$

is surjective and has section $s : \mathcal{N} \mapsto \varphi_{S,M}^{-1}(\mathcal{N}) \leq M$. Here section means: $p \circ s = \text{id}$ for the set on the right.

Exercise. Find an example of $R \rightarrow R'$ flat and M an R -module such that $\{N \leq_R M\} \rightarrow \{\mathcal{N} \leq_{R'} R' \otimes_R M\}$ is not surjective.

Proposition 0.4 (Quotients and localization). Define $\bar{S} := \{s + I \mid s \in S\} \subseteq \bar{R} := R/I$, then:

- (a) As $S^{-1}R$ -modules we have

$$S^{-1}R / S^{-1}I \cong S^{-1}(R/I)$$

- (b) $\bar{S} \subseteq (\bar{R}, 1, \cdot)$ is a submonoid and

$$\psi : S^{-1}\bar{R} \rightarrow S^{-1}\bar{R}, \frac{1}{s}(r + I) \mapsto \frac{r + I}{s + I}$$

is well defined and an isomorphism of rings and of $S^{-1}R$ -modules.

- (c) $S^{-1}I = S^{-1}R \iff I \cap S \neq \emptyset$. In particular $S^{-1}R = 0 \iff 0 \in S$.

Proposition 0.5. Let $M' = R' \otimes_R M$ and $N' = R' \otimes_R N$, then:

- (a) M is a finitely generated R -module $\implies M'$ is a finitely generated R' -module. ($S^{-1}M$ is a finitely generated $S^{-1}R$ -module)
- (b) M is a noetherian R -module $\implies S^{-1}M$ is a noetherian $S^{-1}R$ -module.

Proposition 0.6. Let Q be an R' -module, then:

- (a) The map

$$\begin{aligned} \text{Hom}_R(M, Q_R) &\xrightarrow{\cong} \text{Hom}_{R'}(M', Q) \\ \psi &\mapsto (\psi' : r' \otimes m \mapsto r' \cdot \psi(m)) \end{aligned}$$

is an R' -module isomorphism. In the special case $R' = S^{-1}R$ we get the isomorphism

$$\begin{aligned} \text{Hom}_R(M, Q) &\xrightarrow{\cong} \text{Hom}_{S^{-1}R}(S^{-1}M, Q) \\ \psi &\mapsto \left(\psi' : \frac{m}{s} \mapsto \frac{\psi(m)}{s} \right) \end{aligned}$$

(b) Suppose now that R is noetherian, $\varphi : R \rightarrow R'$ is flat and M is finitely generated as an R -module, then the map:

$$\begin{aligned} \mu_M : R' \otimes_R \operatorname{Hom}_R(M, N) &\xrightarrow{\cong} \operatorname{Hom}_{R'}(M', N') \\ r' \otimes \psi &\mapsto (r'' \otimes m \mapsto r' r'' \otimes \psi(m)) \end{aligned}$$

is an R' -module isomorphism. In the special case $R' = S^{-1}R$ we get the isomorphism

$$\begin{aligned} S^{-1} \operatorname{Hom}_R(M, N) &\xrightarrow{\cong} \operatorname{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N) \\ \frac{1}{s} \psi &\mapsto \left(\frac{1}{s} \psi' : \frac{m'}{s} \mapsto \frac{\psi(m)}{s} s' \right) \end{aligned}$$

(c)

Lemma 0.7. (a) If P is a projective R -module, then $R' \otimes_R P$ is a projective R' -module.

(b) If $P^\bullet \rightarrow M$ is a projective resolution in $\operatorname{Ch}_{\leq 0}^*(R\operatorname{Mod})$ and $R \rightarrow R'$ is flat, then

$$(R' \otimes_R P^i)_{i \geq 0} \rightarrow M$$

is a projective resolution.

(c) If R is noetherian and M is a finitely generated R -module, then there exists a projective resolution of the form

$$\cdots \rightarrow R^{n_2} \rightarrow R^{n_1} \rightarrow R^{n_0} \rightarrow M \rightarrow 0$$

with $n_i \in \mathbb{N}_0$.

(d) If M is a flat R -module, then M' is a flat R' -module.

Theorem 0.8. Suppose $R \rightarrow R'$ is flat and in case (b) that R is noetherian and M finitely generated R -module, then one has natural isomorphisms:

$$(a) \ u_i : R' \otimes_R \operatorname{Tor}_i^R(M, N) \xrightarrow{\cong} R'_i(M', N').$$

$$(b) \ v^i : R' \otimes_R \operatorname{Ext}_R^i(M, N) \xrightarrow{\cong} \operatorname{Ext}_{R'}^i(M', N').$$

0.2 Spectrum of a ring

Definition 0.9. Define

$$\begin{aligned} \operatorname{Spec} R &:= \{\mathfrak{p} \subseteq R \text{ prime ideal}\} \\ \operatorname{Max} R &:= \{\mathfrak{m} \subseteq R \text{ maximal ideal}\} \\ \operatorname{Min} R &:= \{\mathfrak{p} \subseteq R \text{ minimal in } \operatorname{Spec} R\} \end{aligned}$$

Remark. Regarding the definition

(a) $\operatorname{Max} R \subseteq \operatorname{Spec} R$.

$$(b) \ 0_R = 1_R \iff R = 0 \xrightarrow[10]{\iff} \text{Max } R = \emptyset = \text{Spec } R.$$

Theorem 0.10 (Uses Zorn's lemma). *Suppose $R \neq 0$ is commutative, then $\text{Max } R, \text{Min } R$ are non-empty. Reminder on $\text{Min } R$: If $\{\mathfrak{p}_i\}_{i \in I}$ is a descending chain in $\text{Spec } R$, then $\bigcap \mathfrak{p}_i$ is a prime ideal.*

Example. Let R be a PID (not a field), then

$$(a) \ \text{Spec } R = \{(0)\} \cup \underbrace{\text{Max } R}_{= \{(f) \mid f \text{ irr.}\}}$$

(b) Gauß lemma gives the following:

$$\text{Max } R[X] = \{(f, p) \mid p \in R \text{ irr. and } f \in R[X] : \bar{f} \in R/(p)[X] \text{ irr.}\}$$

$$\text{Spec } R[X] = \text{Max}(R[X]) \amalg \{(0)\} \amalg \{(f) \mid f \in R[X] \text{ primitive, irred.}\} \amalg \{(p) \mid p \in R \text{ irred.}\}$$

TODO WTF

Proposition 0.11. *For $\varphi : R \rightarrow R'$ a ring homomorphism, the map*

$$\varphi^* : \text{Spec } R' \rightarrow \text{Spec } R, \mathfrak{p}' \mapsto \varphi^{-1}(\mathfrak{p})$$

is well defined and inclusion preserving.

Proposition 0.12. (a) *For the quotient map $\pi : R \rightarrow R/I (= R')$ we have:*

(i) $\pi^* : \text{Spec } R/I \rightarrow \text{Spec } R$ *is injective.*

(ii) $\text{im } \pi^* = V(I) := \{\mathfrak{p} \in \text{Spec } R \mid I \subseteq \mathfrak{p}\}.$

(iii) $V(I) \rightarrow \text{Spec } R/I, \mathfrak{p} \mapsto \mathfrak{p} + I = \pi(\mathfrak{p}) \subseteq R/\mathfrak{p}$ *is inverse to π^* .*

(b) *For the localization map $\varphi_S : R \rightarrow S^{-1}R, r \mapsto \frac{r}{1}$ we have:*

(i) $\varphi_S^* : \text{Spec}(S^{-1}R) \rightarrow \text{Spec } R$ *is injective.*

(ii) $\text{im } \varphi_S^* = U(S) := \{\mathfrak{p} \in \text{Spec } R \mid \mathfrak{p} \cap S = \emptyset\}.$

(iii) $U(S) \rightarrow \text{Spec}(S^{-1}R), \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ *is inverse to φ_S^* .*

Corollary 0.13 (to Theorem 10). *Suppose $U(S) \neq \emptyset$, i.e. $0 \notin S$, then:*

(a) $U(S)$ *contains minimal and maximal elements with respect to $\subseteq$.*

(b) *If $I \cap S = \emptyset$ then (a) holds for*

$$\{\mathfrak{p} \in U(S) \mid I \subseteq \mathfrak{p}\} = \{\mathfrak{p} \in \text{Spec } R \mid I \subseteq \mathfrak{p}, \mathfrak{p} \cap S \neq \emptyset\}$$

(c) *If $I \subseteq \mathfrak{q}$ for some $\mathfrak{q} \in \text{Spec } R$, then (a) holds for $\{\mathfrak{p} \in \text{Spec } R \mid I \subseteq \mathfrak{p} \subseteq \mathfrak{q}\}.$*

Definition 0.14. For $f \in R$ write

(a) $S_f = \{f^n \mid n \in \mathbb{N}_0\} \leq (R, 1, \cdot)$ *submonoid.*

(b) $U(f) := U(S_f) = \{\mathfrak{p} \in \text{Spec } R \mid f \notin \mathfrak{p}\}.$

(c) $R_f = S_f^{-1}R = \{\frac{r}{f^n} \mid r \in R, n \in \mathbb{N}\}$ *with localization map $\varphi : R \rightarrow R_f$.*

(d) For M an R -module, set $M_f := S_f^{-1}M$.

Note that $U(f)$ is in bijection with $\text{Spec } R_f$ by $\mathfrak{p} \mapsto S_f^{-1}\mathfrak{p} = \mathfrak{p}_f$.

Definition 0.15. $f \in R$ is called *nilpotent* if $\exists n \in \mathbb{N} : f^n = 0$. Note: f is nilpotent $\iff S_f = 0 \iff R_f = 0 \iff U(f) = \emptyset$.

Remark. Recall the definition of a topological space

(a) A topological space is a pair $\underline{X} = (X, \mathcal{T})$ where X is a set, $\mathcal{T} \subseteq \mathcal{P}(X)$ and $\mathcal{T}$ satisfies:

- (i) $\emptyset, X \in \mathcal{T}$.
- (ii) $\forall U, V \in \mathcal{T} : U \cap V \in \mathcal{T}$.
- (iii) $(U_i)_{i \in I}$ in $\mathcal{T}$, then $\bigcup_{i \in I} U_i \in \mathcal{T}$.

(b) Call the elements in $\mathcal{T}$ *open subsets* of X (with respect to the topology $\mathcal{T}$).

(c) Call $A \subseteq X$ *closed* if $X \setminus A$ is open.

(d) If $\underline{X}' = (X', \mathcal{T}')$ is another topological space, call a map $f : X \rightarrow X'$ continuous if $f^{-1}(\mathcal{T}') \subseteq \mathcal{T}$.

Axioms that characterize $\mathcal{A} := \{X \setminus U \mid U \in \mathcal{T}\}$ by:

- (i) $\emptyset, X \in \mathcal{A}$.
- (ii) $\forall A, B \in \mathcal{A} : A \cup B \in \mathcal{A}$.
- (iii) $(A_i)_{i \in I}$ in $\mathcal{A}$, then $\bigcap_{i \in I} A_i \in \mathcal{A}$.

Definition 0.16. Recall also:

(a) A subset $B \subseteq \mathcal{P}(X)$ is called a *basis* for a topology on X if $\forall U, V \in B$ and $x \in U \cap V, \exists W \in B$ such that $x \in W$ and $W \subseteq U \cap V$.

(b) If B is a basis, the topology $\mathcal{T}_B$ generated by B is characterized by: $U \in \mathcal{T}_B \setminus \{X\} \iff \forall x \in U \exists W \in B : x \in W \subseteq U$.

Example. If (X, d) is a metric space, then the set of all ε -balls forms a basis for the associated (metric) topology.

Definition 0.17. A topological space $(X, \mathcal{T})$ is called *quasi-compact* $\iff$ for every covering $(U_i)_{i \in I}$ by open subsets U_i there is a finite $I_0 \subseteq I$ such that $\bigcup_{i \in I_0} U_i = X \iff$ for any family $(A_i)_{i \in I}$ of closed subsets of X such that $\bigcap_{i \in I} A_i = \emptyset$ one finds a finite $I_0 \subseteq I$ such that $\bigcap_{i \in I_0} A_i = \emptyset$.

Theorem-Definition 0.18. Let $\mathcal{A}_{\text{Zar}} := \mathcal{A}_{\text{Zar}, R} := \{V(I) \mid I \subseteq R \text{ ideal}\}$, then:

(a) $\mathcal{A}_{\text{Zar}}$ is the set of closed subset of a topology $\mathcal{T}_{\text{Zar}}$ on $\text{Spec } R$, called the Zariski topology.

(b) The set $B := \{U(f) \mid f \in R\}$ is a basis of $\mathcal{T}_{\text{Zar}}$. Moreover one has

$$U(f) \cap U(g) = U(f \cdot g), \forall f, g \in \text{Spec } R$$

- (c) $(\text{Spec } R, \mathcal{T}_{\text{zar}})$ is quasi-compact.
- (d) If $\varphi : R \rightarrow R'$ is a ring homomorphism, then $\varphi^* : \text{Spec } R' \rightarrow \text{Spec } R$ is continuous with respect to the zariski topologies on $\text{Spec } R$ and R' respectively.

Definition 0.19. For $\mathfrak{p} \in \text{Spec } R$ set

- (a) $S_{\mathfrak{p}} := R \setminus \mathfrak{p}$ submonoid of $(R, 1, \cdot)$.
- (b) $R_{\mathfrak{p}} := S_{\mathfrak{p}}^{-1}R$ localization of R at $\mathfrak{p}$.
- (c) For an R -module M we have $M_{\mathfrak{p}} := S_{\mathfrak{p}}^{-1}M$.

Interpretation: For $\mathfrak{p} \in \text{Spec } R$ let $J := \{U(f) \mid f \in R \setminus \mathfrak{p}\}$

Proposition 0.20 (Exer). (a) J is filtered: $\forall U, V \in J : U \cap V \in J$

(b) For $U(f) \subseteq U(g)$ we have a canonical homomorphism $R_g \rightarrow R_f$.

(c) $R_{\mathfrak{p}} = \text{colim}_J R_f$.

(d) $M_{\mathfrak{p}} = \text{colim}_{f \in J} (M_f)$

Theorem-Definition 0.21. For $R \neq 0$ the following are equivalent:

- (i) $\# \text{Max}(R) = 1$
- (ii) $R \setminus R^{\times}$ is an ideal of R .

If (i) or (ii) hold, R is called a local ring with (unique) maximal ideal $\mathfrak{m}_R = R \setminus R^{\times}$.

Notation. (a) “ $(R, \mathfrak{m})$ is a rocal ring” means R is a local ring with $\mathfrak{m}$ maximal ideal.

(b) The residue field of $(R, \mathfrak{m})$ is $R/\mathfrak{m}$

Exercise 0.22. For $\mathfrak{m} \in \text{Max}(R)$ the following are equivalent:

- (i) $\{\mathfrak{m}\} = \text{Max } R$.
- (ii) $1 + \mathfrak{m} \subseteq R^{\times}$.

Proposition 0.23. Let $\mathfrak{p} \in \text{Spec } R$, then

- (a) $R_{\mathfrak{p}}$ is a local ring.
- (b) $\mathfrak{p}R_{\mathfrak{p}}$